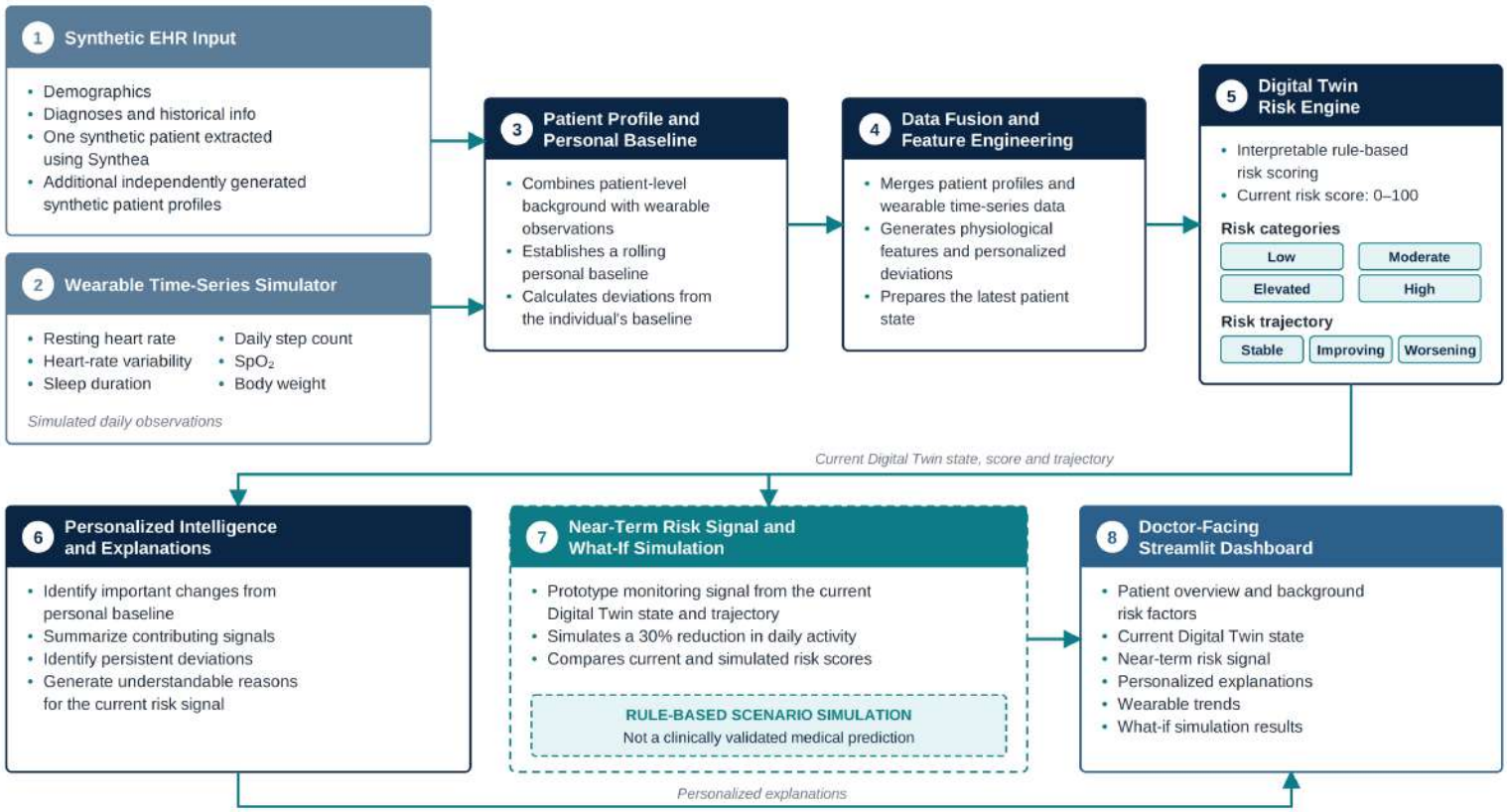
HeartTwin — Personalized Digital Twin for Early Heart-Failure Deterioration

System Architecture · Happiest Health Digital Twin Challenge 2026 · Student-built prototype on synthetic data

Synthetic data sources | Personalized monitoring logic | Near-term signal and simulated scenario | Clinician interface



Side note — ML experiments: Logistic Regression, Random Forest and GRU models were evaluated. The final prototype uses the interpretable rule-based Digital Twin Risk Engine because the tested ML models showed limited performance on the synthetic dataset.

Prototype on synthetic data only. The rule-based risk score is prototype-defined; it is not a validated probability of a future event and is not clinically validated.